

RICKY TANG

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EDUCATION

University of Waterloo

GPA: 3.9/4.0

Honours Bachelor of Software Engineering (Co-op)

Awarded \$11,000+ in scholarships

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Go, C#, C++, Bash, Swift, C, HTML, CSS

Technologies: React, Next.js, FastAPI, Node.js, Express, ASP.NET Core, Flask, RESTful APIs, GraphQL, WebSockets, PyTorch

Tools: Git, Docker, Linux, CI/CD, Redis, MySQL, PostgreSQL, SQLite, MongoDB, ChromaDB, Supabase, AWS, GCP, Prometheus

EXPERIENCE

Hamming AI (YC S24) | TypeScript, Next.js, tRPC, Temporal, PostgreSQL, Prisma

May 2026 – Present

Software Engineer

San Francisco, CA

- Architected **Temporal** signup outreach pipeline with **Exa** research, LLM extraction, **5 lead states** and 4 priority tiers
- Eliminated false timeout warnings for OpenAI STT-only runs via **provider-aware** Temporal external dependency policy
- Built GTM pipeline harvesting **7022** profiles via macOS app-container cache and SQLite mining with **100% field coverage**
- Implemented **schema-invalid** LLM test repair: normalize, dedupe, quality-gate before fallback, backed by 69 unit tests

WAT.ai x Bindwell (YC W25) | Python, FastAPI, DeepEval, Weaviate, TypeScript, React

Jan. 2026 – Present

Software Engineer

Waterloo, ON

- Overhauled async BFS EPA crawler with mime sniffing and retry logic, boosting throughput **59%** and fresh PDF yield **227%**
- Engineered crash-resilient APVMA/FSANZ crawler with SQLite queue, per-prefix auto-blocking, harvesting **3900+** pages
- Architected multi-agent RAG pipeline (intent router, query architect, guardrail) over Weaviate **BM25 + vector** search
- Achieved **1.00 faithfulness** (zero hallucinations) and **1.00 answer relevancy** via DeepEval across 5 golden test questions

Waterloo Aerial Robotics Group (WARG) | Python, React, Flask, OpenCV, MAVLink, Docker

Oct. 2025 – Apr. 2026

Software Engineer

Waterloo, ON

- Streamlined ground station UI with one-click pause/resume for missions, eliminating manual switching for **50+ operators**
- Reduced mission failure recovery time from 30s+ to 3-5s for command pipeline operations (**85% improvement**)
- Engineered full-stack control pipeline with React frontend, Flask-SocketIO backend, and MAVLink for real-time commands
- Implemented OpenCV object detection in aerial imagery and MAVLink telemetry streaming, achieving **80%+ IoU accuracy**

PROJECTS

Tark – Google Earth for Game Devs | TypeScript, Next.js, Python, FastAPI, Redis, Prometheus, Leaflet, SciPy, Numpy, Pytest

- Developed web app turning locations into game-ready 3D meshes in **<15 seconds** (typically weeks of manual modeling)
- Processed Mapbox elevation and satellite imagery to generate terrain meshes at **45K+ triangulated faces per second**
- Extracted **2000+ building footprints** from OpenStreetMap and generated textured .obj files for Unity/Blender workflows
- Productionized with **Redis** job store, reducing latency by **30%** via asyncio workers with **98.2% accuracy** in pytest suites

BrainLattice – PDF to Obsidian Vault Generator | Python, FastAPI, PostgreSQL, Redis, TypeScript, React, AWS, Cloudflare

- Developed React + FastAPI web app and **npm CLI** converting PDFs into Obsidian vaults with NetworkX graph generation
- Reduced token costs by **70-80%** via global concept seeds and batched extraction replacing full-document context caching
- Accelerated processing **70%** via parallelization and embedding deduplication, generating **100+ node graphs** in **<1 minute**
- Built event-driven pipeline using **QStash + Lambda + R2** with **pgvector RAG** for on-demand notes, reaching **300+ installs**

Quota – VS Code Extension for Cost Optimization | TypeScript, VS Code API, LangChain, Python, FastAPI, Next.js, MongoDB, GCP

- Developed VS Code extension analyzing API/cloud/DB overhead with inline annotations and optimization suggestions
- Achieved <7s initial indexing and **<1s** refresh analysis via file hash caching, outperforming AI IDEs by **47x in speed**
- Implemented hybrid AST + regex parser detecting **2x more cost issues** than AI IDEs while saving **~50k tokens per check**
- Architected RAG-powered web sandbox with **LangChain + FAISS** delivering architectural recommendations in **<2s**

AWARDS

- DeltaHacks 2026: **1st Place Overall** | Hack Western 2025: **Best AI Application Built with Cloudflare**
- GenAI Genesis 2026 (Canada's largest AI hackathon): **Top 10 Overall** (out of 250+ teams)